

# Lesson 05 - Algorithms and Pseudocode

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## Exercise 01 - Algorithms

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### Task 01

Below is an algorithm to draw a square of size 50 Units

1. Start at a specific point on the paper.
2. Put Pen down on paper
3. Move forward a 50 Units to draw the first side of the square.
4. Turn right 90 degrees.
5. Move forward the same distance as before to draw the second side of the square.
6. Turn right 90 degrees.
7. Move forward the same distance again to draw the third side of the square.
8. Turn right 90 degrees.
9. Move forward the same distance once more to draw the fourth and final side of the square.
10. Lift up pen
11. Stop.

Repeat this process for the following shapes assuming the same 50 unit measurement

- Equilateral Triangle
- Hexagon
- Circle

### Task 02

Working in Pairs use the online Maze Generator at <https://www.mazegenerator.net/> to create a maze. Download this maze as a pdf or png image or use the maze supplied below.

Create an algorithm to solve this maze starting from the bottom and moving to the exit at the top.

#### Hint

Player can move forward until they meet a wall then they need to turn left or right. The first 2 steps are shown below

1. Player moves forward until they meet a wall
2. turn left
3. move forward
4. turn right

## Exercise 02 - Pseudocode

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### Task 01

Create pseudocode for each of the algorithms you created in Task 01 above. An example of the square pseudocode is shown below

```
Set starting point on the paper  
Put pen down on paper  
Move forward 50 units  
Turn right 90 degrees  
Move forward 50 units  
Turn right 90 degrees  
Move forward 50 units  
Turn right 90 degrees  
Move forward 50 units  
Lift up pen  
Stop
```